

1. Fine-tuning Strategy

In this section, our goal is to identify a suitable base model to serve as *ClassifyLM* and fine-tune it for the task of KA classification using our curated dataset. We detail the baseline methods analyzed (Section 1.1), the evaluation metrics used for comparison (Section 1.2), and the results of our experimental analysis of the candidate models (Section 1.3).

1.1. Baselines

We consider the following NLP methods for the *ClassifyLM* task:

- **Zero-shot Learning:** In zero-shot learning, we simply *prompt* a LLM without domain-specific fine-tuning. We selected DeepSeek-V3 (Liu et al., 2024) and ChatGPT-4o-mini (OpenAI, 2023) for this baseline, since both models are widely available and relatively cost-effective. Additionally, we evaluate Qwen-2.5b-7B-Instruct (Team, 2024) (the model adopted for PreprocessLM), to motivate that different models were used for PreprocessLM and *ClassifyLM*. The specific prompt used for zero-shot classification is detailed in Figure 1 in 4.
- **Random Forest (RF):** RFs are a well-known traditional machine learning method. In contrast to deep learning methods that learn directly from raw text, RFs require feature extraction. Here, we use both Bag-of-Words (BoW) and Term-Frequency-Inverse Document Frequency (TF-IDF), that assign numerical values to words based on how frequently they appear in the training set.
- **LSTM:** To establish a direct comparative baseline with the most relevant related work by Dzurenda et al. (Dzurenda et al., 2024), we utilized the core architecture described in their paper. This model employs the standard BERT (Devlin et al., 2019) tokenizer for text processing, followed by a unidirectional Long Short-Term Memory (LSTM) network for the final classification. Note that it was not possible to use Dzurenda et al.’s trained model directly, as it was optimized for classifying 12 European Union Agency for Cybersecurity (ENISA) work profiles rather than the 9 CSEC2017 Knowledge Areas. Therefore, we adapted their architecture and trained it on our dataset with the recommended learning rate of 0.001 for 20 epochs.
- **Fine-tuned Transformer Models:** We experimented with several transformer variants, namely BERT (Devlin et al., 2019), ALBERT (Lan et al., 2019), RoBERTa (Liu et al., 2019), and DistilBERT (Sanh et al., 2019). BERT serves as a foundational benchmark. DistilBERT offers a more parameter-efficient alternative, reducing the model size by approximately 40% through knowledge distillation. RoBERTa improves upon BERT with optimized training techniques like dynamic masking and extended training, while ALBERT introduces mechanisms to minimize the memory footprint. All models were fine-tuned for a maximum of 20 epochs using a sigmoid output activation and the Binary Cross Entropy loss to accommodate the multi-label classification setting.

1.2. Metrics

The task of assigning course descriptions to one or more KAs is a multi-label, multi-class classification problem. While the accuracy provides an intuitive measure of overall prediction correctness, it is not well-suited for multi-label classification. It is highly affected by class imbalance: models can achieve a high accuracy solely by performing well on the KA with the highest number of samples. Accuracy also treats all errors equally and does not distinguish between false positives and false negatives, even though these could have different implications for curriculum analysis. Therefore, to enable a more nuanced performance evaluation, we do not use the accuracy, but instead compute the macro-averaged precision, recall, and F1 score. Here, ‘macro-averaged’ indicates that metrics are computed per KA and then averaged, such that KAs with fewer samples contribute as much to the final score as KAs with more samples. The definitions of the metrics used are as follows:

- **Precision:** The precision or *true positive rate* measures the proportion of correctly predicted positive labels among all predicted positives for a given class. We report the *macro-averaged precision*, which computes the metric independently for each KA and then takes the mean, assigning equal weight to all KAs. This is defined as:

$$Precision = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i} \quad (1)$$

Here, TP_i represents the number of True Positives (TPs) for KA i , while FP_i stands for the number of False Positives (FPs). C is the total number of classes, which is equal to 9 in the case of KAs. Macro-averaged precision yields a value between 0 and 1, where a high value indicates a low rate of false positives across the model’s predictions.

- Recall: The recall or *sensitivity* quantifies the proportion of actual positives that have been correctly identified as positive by the model. Analogous to precision, we report the *macro-averaged recall*:

$$Recall = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i} \quad (2)$$

Here, FN_i stands for the number of False Negatives (FNs) for a given KA i . The macro-averaged recall yields a value between 0 and 1, where a high recall can be achieved when the model has few false negatives.

- F1 score: The F1 score is the harmonic mean of the precision and the recall, addressing scenarios where one of these two metrics dominates. We report the *macro-averaged F1 score*, calculated as the mean of the per-class F1 scores:

$$F1 = \frac{1}{C} \sum_{i=1}^C \frac{2 TP_i}{2 TP_i + FP_i + FN_i}. \quad (3)$$

The F1 score ranges from 0 to 1, where a higher value indicates superior performance by jointly considering precision and recall. Although the F1 score is the harmonic mean of precision and recall, its macro-average is not. Consequently, the macro-averaged F1 score can be lower than the macro-averaged precision and recall.

1.3. Model Selection

Table 1: The classification performance of various NLP methods on the fine-tuning dataset. Column 1 gives an overview of the NLP model class used, Column 2 provides model specifications. The performance metrics used are the macro-averaged Precision (0–1), Recall (0–1), and F1 score (0–1), as defined in Section 1.2. The reported scores were computed using 10-folds cross validation and have been averaged over 5 random seeds.

Category	Specification	Precision (↑)	Recall (↑)	F1 score (↑)
Zero-shot	Qwen-2.5b-7B-Instruct (Team, 2024)	0.29	0.44	0.32
	ChatGPT-4o-mini (OpenAI, 2023)	0.40	0.33	0.29
	DeepSeek-V3 (Liu et al., 2024)	0.38	0.49	0.41
Random Forest	Bag-of-words	0.67	0.44	0.53
	TF-IDF	0.71	0.44	0.54
LSTM	BERT tokenizer (Dzurenda et al., 2024)	0.14	0.13	0.13
Transformers	ALBERT (Lan et al., 2019)	0.68	0.48	0.55
	DistilBERT (Sanh et al., 2019)	0.71	0.57	0.62
	RoBERTa (Liu et al., 2019)	0.69	0.58	0.63
	BERT (Devlin et al., 2019)	0.71	0.59	0.64

Table 1 gives an overview of the cross-validation scores of all baseline NLP methods mentioned in Section 1.1, as computed on the fine-tuning dataset (described in our main paper). To ensure a robust performance estimate, we use *k-fold cross validation*. This technique partitions the fine-tuning dataset, consisting of synthetic CSEC2017 data and NIST Knowledge Descriptions as described in the previous subsection, into k non-overlapping folds. In each of the k iterations, the model is trained on $k - 1$ folds (corresponding to 90% of the data for $k = 10$) and validated on the hold-out fold (the remaining 10%). In this way, it is possible to compute the performance of the model on the entire fine-tuning dataset. To balance the bias and variance trade-off associated with this process, the recommended setting for k is 10. We adopt $k = 10$ and split the training dataset into 10 non-overlapping 90/10 train/val splits.

We first evaluate zero-shot classification performance using Qwen-2.5b-7B-Instruct (which also serves as the backbone for PreprocessLM), and the state-of-the-art DeepSeek-V3 and ChatGPT-4o-mini models. An identical prompt, as indicated in Figure 1 in 4, was used for all models to ensure a fair comparison. As mentioned in earlier sections, we hypothesized that the tasks designed for PreprocessLM and ClassifyLM cannot be executed by the same model. From the data in Table 1 we can see that the Qwen-2.5b-7B-Instruct model does not have a good off-the-shelf performance for KA classification, a limitation that cannot be remedied via fine-tuning due to the small size of our dataset. The more powerful zero-shot DeepSeek-V3 and ChatGPT-4o-mini models also achieve low scores, demonstrating the limitations of general-purpose LLMs without domain-specific fine-tuning.

Random Forest (RF) based methods showed moderate improvements over zero-shot LLM prompting, which suggests that the features extracted from the fine-tuning dataset possess predictive value. Nonetheless, the limited performance of the RF models indicates that these models struggle to capture the complex semantics inherent

to cybersecurity concepts. The baseline method proposed by Dzurenda et al. (2024), based on a unidirectional LSTM trained on pre-tokenized text, performed significantly worse than all other approaches. However, this can in part be attributed to the fact that the hyperparameter settings and training methodology were not optimized for our dataset, but set to the default values recommended by Dzurenda et al. (2024). A bidirectional LSTM or a deeper neural network architecture could have improved the performance of the LSTM-based approach, but these architectural changes are beyond the proposed methodology of Dzurenda et al. and therefore considered outside of the scope of this paper.

In contrast, Transformer-based architectures substantially outperformed other approaches, with BERT achieving an F1 score of 0.64. We attribute this superior performance to the self-attention mechanism in transformer models, which allows them to weigh the importance of all words in a context simultaneously, capturing long-range contextual information more effectively than LSTMs or RFs. While transformer-based models generally outperformed traditional methods, BERT emerged as the best candidate. Based on these results, we therefore designate the fine-tuned BERT model as ClassifyLM.

2. Constrained Optimization for curriculum composition

After fine-tuning the ClassifyLM model, it can be used to estimate the KA distribution of any given course based on its course description. The model produces a probability distribution over the CSEC2017 Knowledge Areas, providing a representation of the course’s cybersecurity coverage.

The target requirements from the NICE framework for job roles, job categories, and the overall market, are similarly encoded as a probability vector over KAs. This shared vector representation enables the direct alignment of courses with cybersecurity workforce demands, framing the task of optimal course selection as a constrained optimization problem.

To design a curriculum aligned with job market requirements, it is important to consider the structural constraints of the curriculum. In particular, factors such as the total number of credits in the program, and the desired distribution between core and elective credits effectively *constrain* the optimization process. For example, the two-year MSc in Cybersecurity at KTH consists of 120 European Credit Transfer and Accumulation System (ECTS) credits, of which 39.5 ECTS correspond to mandatory core courses, 30 ECTS to conditionally elective courses, 30 ECTS are dedicated to the master thesis project and the remaining 20.5 ECTS are free choice electives. As presented in the case example (in the main body of our paper), CurricuLLM can be used to identify the optimal set of conditional electives for a specific target job role. The program offers 12 elective courses to first-year students, of which students are required to select at least 4. Although CurricuLLM can also be used to recommend free-choice electives, the present scenario focuses on optimizing the selection of conditional electives for simplicity. We formulate the optimization problem for the present scenario as a second-order constrained optimization problem, with the objective of aligning the student’s composed curriculum as closely as possible with the desired KA distribution of the target job role.

Here, we let

- m : Number of KAs, k : Number of elective courses to be selected, n : Total number of available electives, c_M : Total number of mandatory credits.
- c_i the credit attributed to course i .
- $E_{i,j} \in [0, 1]$: Contribution of KA j to elective course i . This is a normalized probability vector, such that $\sum_{j=1}^m E_{i,j} = 1$.
- $T_j \in [0, 1]$: Target KA distribution of the desired job role. This is a normalized probability vector, subject to $\sum_{j=1}^m T_j = 1$.
- $x_i \in \{0, 1\}$: Binary decision variable indicating whether elective i is selected.
- $\mathcal{M} \subset \mathbb{R}^m$: The aggregated KA vector of the mandatory courses.

Such that the MIP formulation becomes

$$\text{minimize} \quad \sum_{j=1}^m \left| \frac{c_M}{c_M + \sum_{i=1}^n c_i x_i} \mathcal{M}_j + \frac{1}{c_M + \sum_{i=1}^n c_i x_i} \sum_{i=1}^n c_i x_i E_{i,j} - T_j \right| \quad (4)$$

$$\text{subject to} \quad \sum_{i=1}^n x_i = k \quad (5)$$

$$x_i \in \{0, 1\} \quad \forall i \in \{1, \dots, n\} \quad (6)$$

Each course (mandatory or elective) is represented as a continuous m -dimensional probability vector indicating the proportional contribution of each KA to the course. Specifically, each elective $i \in 1, \dots, n$ is associated with a normalized contribution vector $E_i = [E_{i,1}, \dots, E_{i,m}]^\top$. The mandatory curriculum component is similarly represented by a vector $M \in [0, 1]^m$ and each elective course $i \in \{1, \dots, n\}$ has a KA contribution vector $E_i \in \mathbb{R}^m$. The desired job profile is represented as a target KA distribution $T \in \mathbb{R}^m$, obtained from Table 3 in (Ramezani and Niemi, 2024). Binary variables $x_i \in \{0, 1\}$ are introduced to indicate whether a given elective course i has been selected. The total number of electives selected is constrained to k using constraints (5) and (6). Additionally, because mandatory and elective courses carry different credit values at the program at KTH, we aggregate KA distributions in a credit-weighted manner. We let c_M denote the total number of mandatory credits and c_i the credits of elective i .

The KA distribution of the student’s entire curriculum is then computed as the sum of the mandatory component, M , and the contributions of the selected electives: $\sum_{j=1}^m \sum_{i=1}^n c_i x_i E_i$, weighted by the number of credits contained in each. The objective (4) is then to minimize the absolute deviation between the KA profile of the student’s curriculum and the target profile T . This ensures that the selected electives fill the knowledge gap left by the mandatory curriculum as closely as possible.

We next examine to what extent this optimization problem should be reformulated to accommodate alternative curriculum constraints. The one-year MSc in Cybersecurity at Nanyang Technological University (NTU) consists of 30 Academic Units (AU), of which 12 AUs are reserved for mandatory courses and 6 AUs are dedicated towards a capstone project. The remaining 12 AUs are conditionally elective, where students choose four elective courses from a pool of nine. Because of the great structural similarity of this program with the program offered at KTH, CurricuLLM can be applied to the NTU master in a similar fashion as mentioned before.

The two-year M.S. in Information Technology - Information Security (MSIT-IS) offered at Carnegie Mellon University (CMU) consists of 75 core units, 36 free program electives, and 12 restricted electives. While the restricted electives can be chosen in a similar way as posed in the optimization formulation for KTH, the core units have a more complicated structure that we consider outside the scope of our current work. The core units are separated into a networking core (12 units, one course selected out of four options), an MSIT-IS core (12 units, one course selected from three options), a Security core (12 units, fixed), a Business and management component (12 units, comprising either two fixed courses or one alternative course), and a practical project worth 24 units. Additionally, we have not analyzed constraints such as the sequentiality of courses (e.g., Advanced calculus requires prior completion of Basic Calculus), practical constraints such as clashing timeslots, and the student’s prior knowledge.

3. Curriculum Analysis extended results

For each of the 3 curricula analyzed in our paper (KTH, NTU, CMU), we randomly selected 3 courses to be analyzed in further detail. Here, the results for the Building Networked Systems Security (KTH) course are provided earlier in the paper in Table 8. The rest of the results can be found in the following tables: Ethical Hacking (KTH), Table 2, Language-Based Security (KTH), Table 3, Network Security (NTU), Table 4, Software Security (NTU), Table 5, Privacy Preserving Technologies & Security in AI (NTU), Table 6, Advanced Real-World Networks (CMU), Table 7, Networked Systems Course (CMU), Table 8, Information Security Policy and Management (CMU), Table 9.

For each course analyzed, we derived topics from the course descriptions using PreprocessLM, labeled them using ClassifyLM and present the final result of this process under the column CurricuLLM in the table. Then, Control Group A, consisting of 34 cybersecurity experts, and a group of three cybersecurity experts previously familiar with the CSEC2017 framework (noted as ‘Expert Group X’) were consulted to independently label the topics. The numbers listed in the tables down below represent these annotations, where the corresponding KAs for each number can be found in the main body of our paper.

4. Zero-shot classification prompt

We employed zero-shot prompting of ChatGPT-mini-4o, and DeepSeek-V3 as a baseline for comparison with the proposed ClassifyLM structure (see Section 1.1). Zero-shot prompting requires the selected model to perform Knowledge Area annotation based solely on the instructions and definitions provided in the prompt, without any prior examples. The prompt provided to both ChatGPT-mini-4o and DeepSeek-V3 is detailed in Figure 1.

Table 2: Topics extracted from the Ethical Hacking Course (KTH) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Ethical Hacking	1,6	6,8	2,5,6	0-8	2,7,8	0,2-6	8
Network and vulnerability scanning	4	3,4,5,7	1,4	4,5	5	0,4	4
Exploit development platforms	2,7	3	2,3	2	2	2,3,4	2
Command and control	0	2	2,3,6	0	7	0,2,4,5	0
Password cracking	1	5,6	1,2,3	0,1	5	1,2,6	6
Independent attack project	1,3	-	7	0	5	0	5
Virtual environment setup	4	7	1,2	4,5	0	0	4

Table 3: Topics extracted from the Language-Based Security Course (KTH) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Language-Based Security	1,6	2	-	0,2	6	0,1,2,6,7	5
Fundamental principles, models and concepts for computer security	0	2	1,4,5,7,8	0-5	1,3	0-8	5
Software security by information flow control	3	4	1,2,4	2	2	2	2
Web application and database security	2	2	1,4,5,6,7	0,2	2,5	1,2,4,5,7	7
Security for mobile applications	5,6	2	1,3,4,5,6	4	2,4	2,5,6	7
Hot topics in computer security	1	-	0,5,8	0-8	1,3,4,5	0-8	7
State-of-the-art in programming language for security	1,6	2	2,3,4	2	2	2	7

Table 4: Topics extracted from the Network Security Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Network Security	5	3,4,5,7	1,2,5,6,7	4	4,5	4	7
Introduction to Networking	5	1,2,3,4,5	4	0	0	4	4
Security in Networks	5	3,4,5,7	1,2,5,6	4	4,5	4	7
Network Programming	5	4,5	2	0,4	2,4	2,4	7
Packet Trace Analysis with Wireshark	5	4	1,4,7	4	4	3,4	4
TLS and Security in Networked Systems	5	1,2,4,5	1,4,5	4	4,5	4	4

5. KA distribution NICE Job Roles

The relative importance of each of the nine Knowledge Areas for all work role categories defined in the NICE framework v2.0.0 is shown in the main body of our paper. The ‘relevance’ of KAs was derived by collecting the KDs associated with a given job role, annotating them with corresponding KAs using ClassifyLM, and computing the resulting distribution of KAs. For completeness, this appendix provides the relative importance of the KAs for *all* individual NICE work roles. These detailed results are displayed in Table 10.

6. Knowledge Area 0: Miscellaneous

In (Ramezani and Niemi, 2024) the authors identified seven knowledge units in the knowledge area 0: Miscellaneous. The description of these knowledge units are as follows.

- *Computer Science*: includes topics related to Programming and Software Development, System Architecture, Database Management, Mathematics, Information Systems, and Human and Organizational Aspects of Information Technologies.

Table 5: Topics extracted from the Software Security Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Software Security	3,4,5,7	–	2	2	2	2	2
Challenges in software security	1–5	8	2	2	2	2	2
Principles of secure software development	3–5,7	2	0,2,5	2	2	2	2
Mechanisms for secure coding	4	2,5	5	2	2	2	2
Tools for software security	2,4	1	2	2	2	2	2
Causes of vulnerabilities	1,2,4,5	3,6	0,6,7	0–6	2–5	0–8	7
Strategies to avoid vulnerabilities	1–3	2	3,5,7	0–6	2,3	2,5,6,7,8	7
Defenses against attacks	1–5,7	1,6	3,5,7	0–6	5,7	0–8	5
Secure programming practices	2	3	2,3,5,6,8	2	2	2	2
Language-specific secure coding	2,4	3	2,3,5,7	2	2	2	2
System-level defenses	1,2,4,5	5	1–5	5	5	5	5
Architectural approaches for security	1,3–5,7,8	2,4	0,3–8	4	5	0,5,6,7,8	2
Run-time enforcement techniques	2–6	1,5	0,4,6–8	4	2,3	2,3,5	7
Software analysis tools	1,2,4,6,7	2	0–2,4–8	2	0,2	0,2,3	2
Vulnerability detection methods	1–3,5–8	3	0,2,5–8	3	5	0,2–6	2,3
Application of security tools in scenarios	0,2,5–8	–	0–2,5–8	3	5	0	7

Table 6: Topics extracted from Privacy Preserving Technologies & Security in AI Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Privacy Preserving Technologies & Security in AI	1,6	0–3,5,7,8	0–2,6–8	0,2,3,6	5,6,8	0,1,6	1,6–8
Fully Homomorphic Encryption	1	0,1,4	1,3,5	1	1	1	1
Secure Multi-computation	1,2,4	0,2,3,5,7	1,3,5,7	1	1	1	1
Searchable Encryption	1	0,1,4	1,2,5	1	1	1	1
Data Sharing in Machine Learning	1	0,4,7,8	0–2,4,5,7	0,1	0	1,6	4

Table 7: Topics extracted from Advanced Real-World Networks Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Advanced Real-World Networks	4,5	1–8	1,4,5	0,4	0	4	4
4G and 5G network infrastructures	4	1–8	0,3–5	0,4	0	1,4,5	4
IPv6	2,5	0	0,1,4	4	0	4	4
SDN and VFN	2	1,2,4,7	0,1,2,3,5	4	0	2,4	4
Data centers	1	1,3–6	0–7	0	0	1–5	1
Mesh networks	3,4	1–5	0,4	4	0	4	4
Embedded networks	3,4	1–8	0,4	4	0	2,4	4

- *Business and Law*: includes topics related to Regulatory Frameworks, Organizational Operations & Policies, Stakeholder Relationships, and Management & Human Resources.
- *Communication and Networking*: includes topics related to Telecommunications Fundamentals, Network Architectures Fundamentals, Network Management, Communication Systems, and Information Gathering and Dissemination.
- *Information Technology*: includes topics related to Technology Infrastructure, Content Systems, Technology Trends & Integration, and Media, Communication, & Dissemination.
- *Cyberspace Practice*: includes topics related to Adversarial Models, Cyber Attacks & Vulnerabilities, Cyber Operations & Practices, Core Cyber Concepts & Principles, and Security Tools, Vendors, & Defensive Technologies.

Table 8: Topics extracted from Security in Networked Systems Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Security in Networked Systems	1	1–5	1	4	4,5	4	7
Network and transport-layer attacks and defenses	4	1–5	4	4	4,5	4	4
Network intrusion detection	5,6,7	1–5	1,4	4	4,5	4	4
Denial of service (DoS) and distributed denial-of-service (DDoS) detection and reaction	4,5,8	4	1,4	4	4,5	4	4
Worm and virus propagation	2,3	6	2,3,5,7	4	2,4,5	0,2–6	1
Tracing the source of attacks	0–8	0–2,4–6–,8	1,2	0–6	5	0,2,4,5	1
Traffic analysis	4	0,1,4,8	4	4	1,4	0,4	7
Techniques for hiding the source or destination of network traffic	1,2,5	0,1,4,6,8	2,4,5	4	4	4	4
Secure routing protocols	4	0,1,2,4,8	4	4	4	4	4
Content poisoning attacks	1,2,3	1–4	1	4	2–4	2,4	1
Advanced techniques for reacting to network attacks	4,8	0–6	0,6–8	4	4,5	0,4	4

Table 9: Topics extracted from Information Security Policy and Management Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Information Security Policy and Management	0,1,5,6,8	7,8	7	7	5,7	0,7,8	7
Security marketplace overview	0,2,8	3	0	7	5,7	0	7
Decision making with multiple parties involved	5,7,8	6,8	8	7	0	0,1,7,8	7,8
Role of policy in information security	1,7	8	7	7,8	5,7,8	7	7,8
Intra-organization policies	4,7	4,8	3,5,6	7	0	7	7
Market and competition impact on security provision	0,7,8	1,2	8	7	5,7,8	0,7,8	7
Causes of market failure (externalities)	0,5,7,8	0,7	6–8	0,7	0	0,7,8	7
Policy tools to mitigate market failure	0,5	0,8	3,7,8	7	0	0,7	7,8
Key laws and regulations on product liability and security standards	5	3,7,8	1,2	7,8	8	0,3,4,5,7,8	7
Overview of security industry trends, technologies, vendor and user strategies	0	0–5	0,4	7	8	0	7
Managerial and policy issues in information security provision	5,7	0,1,3,6	0,2	7	7,8	7	7
When and how policy intervention is needed	–	0,7,8	6,7	7,8	8	0,7,8	7,8

- *Pedagogy*: includes topics related to Learning Frameworks, Educational Technology, Training Policies, Organizational Education Practices, Assessment Techniques, Experiential & Applied Learning, and Psychology, Behavior, & Human Factors.
- *Intelligence*: includes topics related to Intelligence Fundamentals, Intelligence Frameworks, Intelligence Collection, Intelligence Management, Targeting & Operational Intelligence, Intelligence Production, Intelligence Dissemination, Intelligence Planning, Resource Prioritization, Partner & Organizational Intelligence Integration, and Covert & Specialized Intelligence Techniques.

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Prompt Structure for Knowledge Area Classification

System Prompt:

"You are a helpful AI assistant.

Instructions:

- a. Carefully read the knowledge statement.
- b. Choose one or more of the following (0, 1, 2, 3, 4, 5, 6, 7, 8).
- c. Do NOT include any explanation or additional text in the response."

User Message:

"#Question: Classify the following statement {knowledge} into one or multiple of the following knowledge areas:

Options: {"0": "miscellaneous (this includes Computer Science, Business and Law, Communication and Networking, Information Technology, Cyberspace Practice, Pedagogy, and Intelligence)", "1": "data security", "2": "software security", "3": "component security", "4": "connection security", "5": "system security", "6": "human security", "7": "organizational security", "8": "societal security"}"

Figure 1: The prompt template used for annotating knowledge statements with Knowledge Areas using Zero-Shot prompting of DeepSeek-V3 or ChatGPT-4o-mini. The placeholder {knowledge} is replaced by a NICE Knowledge Description or a synthetically derived CSEC2017 topic.

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Table 10: Relative importance of CSEC2017 Knowledge Areas for all Work Roles and Work Role Categories contained in the NICE Framework v2.0.0. Knowledge Descriptions contained in each work role were annotated using CurricuLLM and the numbers represent percentage-wise contributions, rounded to the nearest integer.

Work Role	KA0	KA1	KA2	KA3	KA4	KA5	KA6	KA7	KA8
OVERSIGHT and GOVERNANCE (OG)	16	7	3	1	9	3	7	41	12
Communications Security (COMSEC) Management	10	12	5	5	5	4	8	38	11
Cybersecurity Policy and Planning	15	6	2	0	8	2	6	44	17
Cybersecurity Workforce Management	19	4	3	0	7	0	4	51	12
Cybersecurity Curriculum Development	33	6	0	0	11	2	8	29	12
Cybersecurity Instruction	26	8	0	0	17	3	8	26	10
Cybersecurity Legal Advice	17	12	2	0	5	0	12	36	17
Executive Cybersecurity Leadership	8	5	2	0	15	3	5	46	15
Privacy Compliance	10	11	1	1	11	4	13	31	18
Product Support Management	18	4	4	3	7	3	4	48	10
Program Management	16	5	5	3	8	3	5	45	11
Secure Project Management	18	4	4	3	7	3	4	45	10
Security Control Assessment	23	9	6	3	11	5	6	31	7
Systems Authorization	15	13	3	0	8	5	6	39	11
Systems Security Management	14	7	6	4	12	6	5	39	7
Technology Portfolio Management	7	7	2	0	9	0	7	51	16
Technology Program Auditing	16	6	2	0	6	0	6	51	14
DESIGN and DEVELOPMENT (DD)	14	6	10	4	11	9	5	32	8
Cybersecurity Architecture	17	7	9	4	16	13	5	25	5
Enterprise Architecture	14	6	7	5	14	13	6	28	6
Secure Software Development	13	3	19	4	11	11	4	28	9
Secure Systems Development	17	8	12	3	13	8	6	26	7
Software Security Assessment	12	3	19	5	10	9	4	29	9
Systems Requirements Planning	14	9	6	4	8	8	8	34	9
Systems Testing and Evaluation	14	6	9	5	8	9	6	34	10
Technology Research and Development	18	8	8	5	16	7	3	25	10
Operational Technology (OT) Cybersecurity Engineering	5	7	0	0	7	5	5	62	9
IMPLEMENTATION and OPERATION (IO)	20	11	4	2	12	5	8	29	10
Data Analysis	20	13	7	1	7	8	8	25	11
Database Administration	22	17	0	0	7	1	12	29	12
Knowledge Management	31	8	2	0	8	0	8	31	13
Network Operations	20	10	4	4	21	5	7	23	8
Systems Administration	12	8	5	4	21	7	6	28	8
Systems Security Analysis	18	9	6	4	14	8	6	28	7
Technical Support	18	10	1	0	9	4	10	37	10
PROTECTION and DEFENSE (PD)	15	11	3	1	16	5	6	33	9
Defensive Cybersecurity	17	9	6	3	20	8	5	26	6
Digital Forensics	14	23	5	4	11	9	3	22	10
Incident Response	10	10	1	0	19	3	10	37	10
Infrastructure Support	11	11	0	0	22	3	9	34	11
Insider Threat Analysis	15	10	0	0	8	3	5	51	8
Threat Analysis	32	5	2	0	18	3	5	26	8
Vulnerability Analysis	9	10	4	0	14	6	9	36	14
INVESTIGATION (IN)	12	21	3	2	9	7	5	28	12
Cybercrime Investigation	12	18	1	0	9	4	6	35	15
Digital Evidence Analysis	13	25	5	4	10	10	4	21	10